

Geometry–Dynamics Alignment

Why the Dominant Mode Need Not Be the Important Mode

Bond Repricing Research Stack

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Abstract

High-dimensional systems are often summarized by their largest eigenvalue, largest principal component, dominant factor, or leading Lyapunov mode. This practice hides an assumption: the dynamically dominant direction is also the geometrically important direction. The assumption is not generally warranted. This technical note develops a framework for comparing local geometric importance, represented by Hessian directions of a stability landscape, with dynamical importance, represented by finite-time Lyapunov modes of the realized trajectory. The central object is not a single mode but the full alignment spectrum

$$\mathcal{A} = (a_1, \dots, a_d), \quad a_i = \frac{|\langle v_H, q_i \rangle_M|}{\|v_H\|_M \|q_i\|_M}.$$

It defines the basic invariance, null-geometry, subspace, and spectral-stability structure behind this object, then uses U.S. Treasury-market data from 2020–2026 as a case study. In that application, the leading Lyapunov mode is often an unreliable proxy for Hessian alignment: the strongest mode changes across bandwidths and robustness specifications, while the full spectrum remains the meaningful diagnostic. The bond market illustrates a broader methodological claim: geometric and dynamical importance need not coincide, so principal-mode summaries can be unstable and misleading.

1 Introduction: The Hidden Assumption

Across scientific domains, the first instinct is to look at the largest object: the largest eigenvalue, the first principal component, the dominant factor, the top Lyapunov exponent, or the leading singular vector. This habit is useful when the leading object is known to govern the relevant risk. It is dangerous when it silently equates two different meanings of importance.

This note separates two questions:

Which directions are geometrically important?

Which directions are dynamically important?

The first question concerns the shape of a local landscape. The second concerns how perturbations actually propagate under the observed dynamics. The central question is whether these two objects point in the same directions.

The one-sentence thesis is:

The dominant mode of a high-dimensional dynamical system is generally an unreliable proxy for the alignment between local geometry and realized instability; the relevant object is the full alignment spectrum together with its robustness and null-geometry structure.

The empirical case is a six-dimensional bond-market and monetary-policy state. That case matters, but it is not the main character. The main character is the geometry–dynamics alignment framework.

A concrete failure mode is easy to state. Suppose a bond-risk dashboard watches only q_1 because it is the leading dynamical mode. If the current Hessian instability is almost orthogonal to q_1 but strongly aligned with q_4 , the dashboard can report low geometric risk exactly when the local curvature is pointing into a different propagation channel. The practical question is not “which mode is largest?” but “where does the unstable geometry project?”

2 Geometry and Dynamics Are Different Objects

Let $x \in \mathbb{R}^d$ denote a standardized system state. A local stability or loss landscape is a scalar field

$$V : \mathbb{R}^d \rightarrow \mathbb{R}, \quad x \mapsto V(x).$$

Its second-order local geometry is the Hessian

$$H(x) = \nabla^2 V(x).$$

If $H(x)$ has an eigenpair (μ_j, e_j) with $\mu_j < 0$, then the landscape is locally concave along e_j . The most unstable Hessian direction is the Rayleigh-quotient minimizer

$$v_H(x) \in \arg \min_{\|v\|_M=1} v^\top H(x) v,$$

where $M \succ 0$ is the metric induced by the chosen standardization and $\|v\|_M^2 = v^\top M v$. When several negative eigenvalues are clustered, the right geometric object is an unstable subspace rather than a single vector.

Dynamics are defined by a different operator. In discrete time, suppose

$$X_{t+1} = F_t(X_t) + \varepsilon_{t+1}, \quad J_t = DF_t(X_t).$$

For horizon h , the tangent map is

$$\Phi_{t,h} = J_{t+h-1} \cdots J_{t+1} J_t.$$

Finite-time Lyapunov exponents are obtained from the singular values of $\Phi_{t,h}$,

$$\lambda_i(t, h) = \frac{1}{h} \log \sigma_i(\Phi_{t,h}), \quad \sigma_1 \geq \cdots \geq \sigma_d,$$

or numerically through a Benettin/QR product [Benettin et al., 1980, Oseledets, 1968, Arnold, 1998]. The associated forward Lyapunov/Oseledets modes are denoted $q_1(t), \dots, q_d(t)$, ordered by dynamic importance.

Nothing in these definitions requires v_H to equal q_1 . They coincide in a special benchmark: gradient flow. If

$$\dot{X} = -\nabla V(X),$$

then, up to linearization around the point being analyzed, the local dynamical Jacobian is $J = -H$, so Hessian curvature and local dynamical propagation are mechanically linked. But the empirical systems of interest here, including monetary-policy and bond-market data, are not assumed to be gradient flows. The alignment must be tested.

3 Alignment as the Central Object

Let $v_H \neq 0$ be a Hessian-geometric direction and let $q_i \neq 0$ be the i th dynamical mode. The metric-normalized mode alignment is

$$a_i = \frac{|\langle v_H, q_i \rangle_M|}{\|v_H\|_M \|q_i\|_M}.$$

The absolute value quotients out the sign indeterminacy of eigenvectors and Lyapunov vectors.

The alignment spectrum is

$$\mathcal{A}(v_H, Q) = (a_1, \dots, a_d), \quad Q = [q_1, \dots, q_d].$$

The alignment spectrum, not a single mode, is the object of interest.

When $(q_1, \dots, q_d)$ is an M -orthonormal frame and $\|v_H\|_M = 1$, the squared alignments

$$p_i = a_i^2$$

are projection masses and satisfy $\sum_i p_i = 1$. Thus $\mathcal{A}$ is the analogue of a vote distribution: a_1 is one coordinate, $\arg \max_i a_i$ is a selected winner, and $(p_1, \dots, p_d)$ is the distribution of geometric mass over dynamical directions.

Two simple checks fix the scale of the statistic. For all nonzero $v, q \in \mathbb{R}^d$ and $M \succ 0$,

$$0 \leq \frac{|\langle v, q \rangle_M|}{\|v\|_M \|q\|_M} \leq 1.$$

The range follows from Cauchy–Schwarz in the inner product $\langle v, q \rangle_M = v^\top M q$. Sign invariance follows from the absolute value. If $U^\top M U = M$, then the value is also invariant under the isometry $(v, q) \mapsto (Uv, Uq)$ because the isometry preserves both the inner product and the induced norm.

The statistic is not invariant under arbitrary invertible reparameterizations. Changing the scaling of state variables changes M , and therefore changes the meaning of angle. Every alignment claim must declare the metric or standardization.

4 Why Principal-Mode Thinking Fails

The principal-mode diagnostic is

$$D_1 = a_1.$$

It is appealing because it is one-dimensional, easy to display, and easy to explain. It is also insufficient whenever the geometric direction has most of its mass away from q_1 .

The insufficiency is immediate. In any dimension $d \geq 2$, take $v_H = q_k$ for some $k \neq 1$ in an orthonormal frame. Then $a_k = |\langle q_k, q_k \rangle| = 1$ while $a_1 = |\langle q_k, q_1 \rangle| = 0$. A dashboard that reports only D_1 can say “no alignment” while the geometric direction is exactly aligned with a subdominant dynamical mode. Conversely, a selected-mode statistic

$$D_{\max} = \max_i a_i, \quad m^* = \arg \max_i a_i$$

can overstate confidence if the identity m^* is unstable across bandwidths, samples, or dynamics estimators. The framework therefore reports the entire spectrum, not just a_1 or m^* .

5 Full-Spectrum Summaries

The spectrum can be summarized without collapsing it to one selected mode. Given weights $w_i \geq 0$ with $\sum_i w_i = 1$, define

$$W(\mathcal{A}; w) = \sum_{i=1}^d w_i a_i.$$

Useful choices include uniform weights, energy weights, and exponent-based weights such as

$$w_i^{|\lambda|} = \frac{|\lambda_i|}{\sum_j |\lambda_j|}.$$

In the current bond-market run, the eigenvalue-weighted alignment is 0.397, the energy-weighted alignment is 0.416, and the uniform alignment is 0.261. The difference is substantive: weighting by dynamical scale raises the spectrum-level read by roughly half relative to the unweighted average, so the alignment is not merely a diffuse background effect. At the same time, the selected q_4 alignment of 0.621 is not the whole story; material mass also sits in other non-principal modes.

These aggregates should not replace the table of a_i values. Their purpose is to prevent the selected mode from becoming the only reported object.

6 Robustness as a Mathematical Object

Let $s \in \mathcal{S}$ index a specification: bandwidth, state variables, target definition, sample window, standardization, and dynamics estimator. Each specification produces

$$\mathcal{A}_s = (a_{s1}, \dots, a_{sd}), \quad m(s) = \arg \max_i a_{si}.$$

The robustness object is the map

$$s \mapsto \mathcal{A}_s,$$

not just one realization. Mode identity is stable only if $m(s)$ is stable over the relevant grid. A compact instability diagnostic is the empirical distribution

$$\hat{p}_k = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \mathbf{1}\{m(s) = k\}.$$

If $\hat{p}_1$ is small, a principal-mode summary is not representative. If $\hat{p}_k$ is spread across many k , exact mode identity is itself unstable.

In the bond-market multiverse, 468 specifications succeed out of 504. The principal mode is strongest in 22.2% of successful specifications, while a non-principal mode is strongest in 77.8%. The strongest-mode counts are

$$(104, 79, 67, 97, 75, 46)$$

for modes $q_1, \dots, q_6$. Mode 4 is strongest in 97 of 468 specifications, but q_1 is strongest in 104 and the remaining mass is spread across the other four modes. This is the empirical reason the note is a framework note rather than a “mode 4” note: the winning index moves; the spectrum is the stable object.

7 Empirical Testbed: Six-Dimensional Bond-Market State

The empirical testbed is a six-dimensional standardized state with policy-rate, inflation/credibility, credit-spread, equity/market, and growth components. The data combine Treasury yields, effective federal funds rates, macro stress panels, equity series, and the full yield curve. The current multiverse run covers 1,600 daily observations from 2020-01-02 through 2026-06-01.

The diagnostic protocol is:

1. Estimate a local stability landscape $\hat{V}(X)$.
2. Compute $H = \nabla^2 \hat{V}$ and extract v_H or an unstable Hessian subspace.
3. Estimate finite-time Lyapunov modes $q_1, \dots, q_d$ from the realized state trajectory.
4. Compute the alignment spectrum $\mathcal{A} = (a_1, \dots, a_d)$.
5. Repeat across bandwidths, samples, targets, state sets, and dynamics estimators.

The baseline alignment spectrum at bandwidth 1.0 is:

Table 1: Baseline Hessian–Lyapunov alignment spectrum at bandwidth 1.0

Mode	Lyapunov exponent	$ \cos(v_H, q_i) $
q_1	−0.00177	0.021
q_2	−0.00177	0.052
q_3	−0.00812	0.083
q_4	−0.01460	0.621
q_5	−0.03434	0.357
q_6	−0.05717	0.432

8 Bandwidth, Smoothers, and Mode Migration

Bandwidth changes the estimated Hessian and can change the location of the unstable geometric direction. Therefore bandwidth is not a nuisance detail; it is part of the reported result.

Table 2: Bandwidth sensitivity of selected alignments

Bandwidth	a_1	a_2	a_4	Hessian eigenvalue
0.50	0.512	0.849	0.007	−161.9
0.75	0.015	0.054	0.537	−55.3
1.00	0.021	0.052	0.621	−73.6

At bandwidth 0.50, alignment lives in the first two modes. At bandwidths 0.75 and 1.00, the leading mode is nearly orthogonal while q_4 becomes a stronger witness. Across the bandwidth grid $\{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$, mode-4 alignment has mean 0.411, standard deviation 0.221, minimum 0.007, and maximum 0.621.

One plausible interpretation is timescale sensitivity. A narrow bandwidth makes the Hessian more local and reactive, so it can emphasize high-frequency curvature near the current state; in this run that curvature aligns with q_1 – q_2 . Wider bandwidths smooth over a broader neighborhood

and can expose lower-frequency or off-principal structure; here that structure appears in q_4 . This should be read as a sensitivity warning rather than proof that q_4 is the “true” channel. Bandwidth, conditioning, and eigengaps all need to be reported before naming a stable mode.

9 Null Geometry

Alignment needs a reference geometry. The Haar null used here is a reference orientation, not a data-generating model. It asks what alignment would look like if the Hessian direction were randomly oriented relative to the declared Lyapunov frame under the chosen metric. The null question is:

What alignment would be expected from geometry alone if the Hessian direction were isotropically random relative to the Lyapunov frame?

Let u be Haar-uniform on the unit sphere S^{d-1} and let q be any fixed unit vector. Then

$$Z = |\langle u, q \rangle|^2 \sim \text{Beta}\left(\frac{1}{2}, \frac{d-1}{2}\right).$$

Thus a fixed-mode tail is

$$\mathbb{P}(Z \geq z) = 1 - I_z\left(\frac{1}{2}, \frac{d-1}{2}\right),$$

where I_z is the regularized incomplete beta function. This is the correct law for a predeclared fixed mode such as q_1 .

Selected-mode claims require a different null. In an orthonormal basis,

$$(|\langle u, q_1 \rangle|^2, \dots, |\langle u, q_d \rangle|^2) \sim \text{Dirichlet}\left(\frac{1}{2}, \dots, \frac{1}{2}\right).$$

The selected statistic

$$M_d = \max_i |\langle u, q_i \rangle|^2$$

must therefore be compared to the max-over-modes null, or at least to a multiplicity-adjusted fixed-mode tail. In the current implementation, the selected tail is estimated by Monte Carlo and Bonferroni fixed-mode tails are recorded as a conservative reference.

For $d = 6$, the fixed-mode alignment is 0.163 with analytic fixed-mode tail 0.727 and Monte Carlo tail 0.715. The selected-mode alignment is 0.663 with max-over-modes Monte Carlo tail 0.623. These numbers are not evidence of a market-isotropy theorem. They are a diagnostic reference: geometry alone can produce nontrivial alignments, and selected-mode reads must pay a selection cost.

The GOE connection is limited. GOE eigenvectors are Haar-oriented under rotational invariance, so the Haar null is a useful reference orientation. But this is not a GOE spectrum or outlier test, and it does not require the market data to be generated by a GOE model.

10 Spectral Stability and Subspace Alignment

Single eigenvectors are meaningful only when the relevant eigengaps are meaningful. If two Hessian eigenvalues or two Lyapunov exponents are nearly tied, the individual mode labels can rotate under small perturbations even when the subspace is stable. Since this note does not report a data-specific eigengap analysis, exact mode labels should be treated as descriptive, not gap-certified.

The framework response is subspace alignment. Let $U \in \mathbb{R}^{d \times p}$ span a Hessian-unstable subspace and let $Q \in \mathbb{R}^{d \times r}$ span a dynamical subspace. The principal angles satisfy

$$\sigma_j(U^\top M Q) = \cos \theta_j, \quad j = 1, \dots, \min(p, r),$$

after M -orthonormalization. Scalar summaries include

$$A_{\text{sub}}(U, Q) = \|P_U P_Q\|_2 = \cos \theta_1$$

and the projection distance

$$d_{\text{Gr}}(U, Q) = \|P_U - P_Q\|_2.$$

The practical response is to report both single-mode alignment and subspace alignment when modes are close. If mode identity moves because gaps are small, the right diagnostic is eigenspace alignment rather than a hard single-index label.

11 Predictive Exploration

The predictive layer is preliminary and subordinate to the framework. It asks whether alignment summaries add forward information for future stress after a baseline information set Z_t is already included:

$$\mathbb{P}(Y_{t+h} = 1 \mid Z_t, \mathcal{A}_t) = \sigma(\alpha + \beta^\top Z_t + \gamma^\top g(\mathcal{A}_t)),$$

where $g(\mathcal{A}_t)$ may include a_1 , D_{max} , weighted alignment, or lagged alignment. The point is not to establish a production forecasting model. It is to check whether the spectrum carries information that the principal mode alone would miss.

A distributed-lag check at horizon 21 with lags 0–10 gives $R^2 = 0.135$, cumulative effect -0.142 , and concentration ratio 0.234. That is best read as a preliminary sign that alignment may have medium-horizon structure, not as a validated propagation law.

The horizon sweep supports a medium-horizon pattern, not a characteristic period. ΔAUC is largest at 63 days, but ΔR_{OOS}^2 and Brier improvement peak at 28 days; at 63 days two of three metrics do not improve.

Table 3: Metric-dependent horizon maxima

Metric	Best horizon	Value at best horizon	Value at 63d
ΔAUC	63d	+0.0181	+0.0181
ΔR_{OOS}^2	28d	+0.0265	−0.0029
ΔBrier	28d	−0.0050	+0.0003

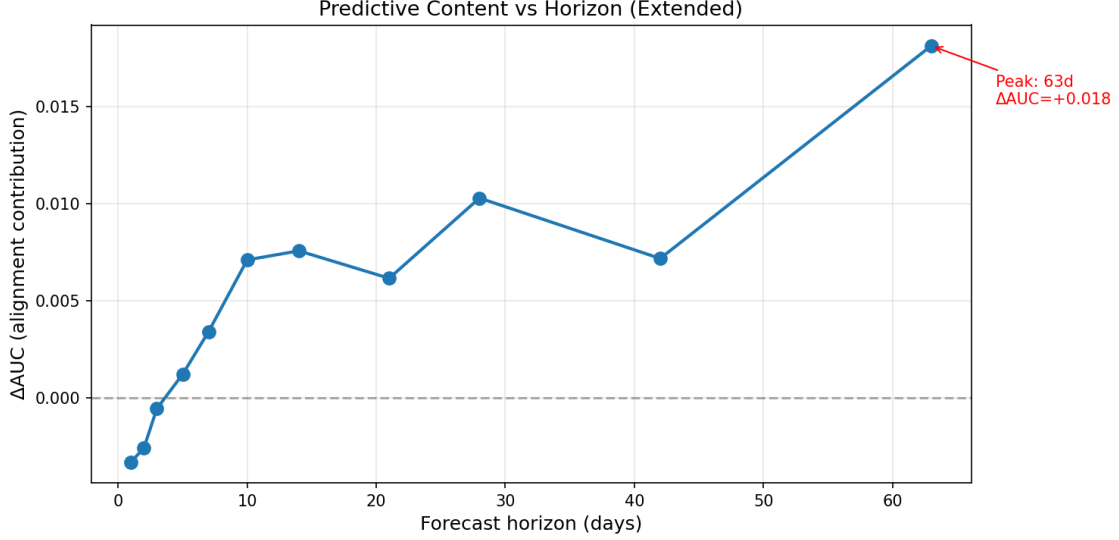


Figure 1: Horizon sweep. The evidence supports medium-horizon incremental content, not a single validated propagation period.

12 Generalization Beyond Finance

The framework is not specific to bonds:

- **Control.** Cost or Lyapunov-function curvature can point in a different direction from closed-loop perturbation growth.
- **Robotics.** Policy or value-function geometry can identify locally fragile directions that do not match the unstable rollout modes.
- **Representation learning.** Hessian or gradient feature importance can differ from training-trajectory drift or recurrent-state dynamics.
- **Scientific machine learning.** Dominant energetic modes in a reduced-order model need not be the modes that matter for a target scientific quantity.
- **Neuroscience and physics.** High-variance population modes can differ from task-relevant or instability-relevant geometric directions.

13 Conclusion

The original note asked whether the biggest mode is the baddest. The stronger technical question is how to compare geometric importance with dynamical importance in high-dimensional systems.

The answer is not a fixed mode index. It is the alignment spectrum

$$\mathcal{A} = (a_1, \dots, a_d)$$

together with its robustness map, null-geometry benchmark, and spectral stability caveats. In the bond-market case, the principal mode is often a poor proxy for Hessian alignment, and the identity of the strongest mode changes across specifications. That is precisely why the framework is needed.

Bottom line: geometric and dynamical importance need not coincide. Principal-mode summaries can therefore be unstable and misleading. The relevant object is the full alignment spectrum together with its robustness and null-geometry structure.

References

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